Supraspinal motor systems: Basal Ganglia

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Make.
Heal.
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Reinvent the Future.

Session Objectives

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1) Identify the nuclei and basic divisions of the basal ganglia
2) Describe the internal and external connections of the basal ganglia including neurotransmitters
3) Understand how the direct and indirect pathways promote and inhibit behavior, and the impact of neuromodulators
4) Understand exemplar dysfunctions and their relation to the basic basal ganglia circuit

Nolte's The Human Brain: An introduction to its functional anatomy
Chapter 19 The Basal nuclei
http://arktos.nyit.edu/login?url=https://www-clinicalkey-com.arktos.nyit.edu/#!/content/book/3-s2.0-B9780323653985000199

General features of the basal ganglia

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On
old
Olympus's
towering
top
a
Finn
and
German
viewed
some
hops

- Collection of gray matter nuclei: some surrounding the thalamus, others in the midbrain.
- Communicate with diverse cortical regions via the thalamus.
- Involved with selecting automated actions and thoughts.

Cortical loops through the basal ganglia*

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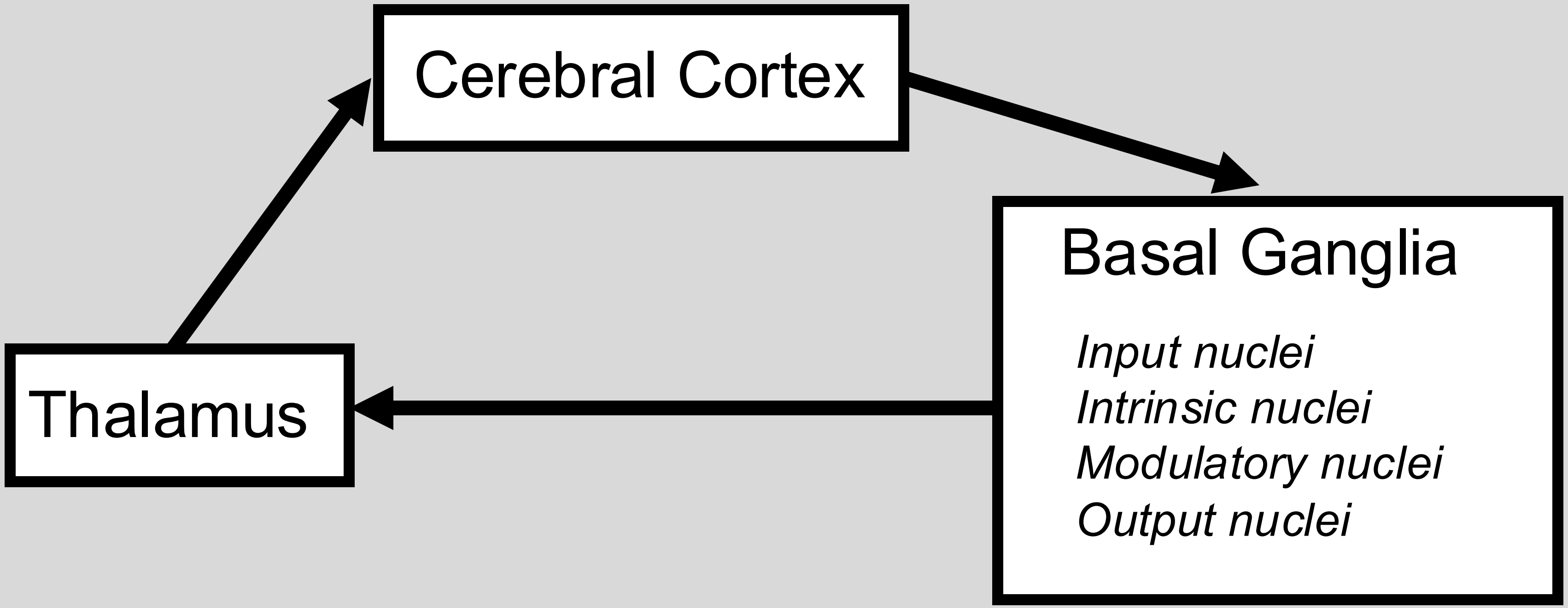

Which nuclei are which?
How do they communicate among themselves?
What neurotransmitters does each release?
How does all this create organized behavior?

Basal ganglia communicates with ipsilateral cerebral cortex via ipsilateral thalamus

Ten nuclei of the basal ganglia

Striatum

3

Pallidum

3

Nuclei within Midbrain and Diencephalon

4

Most release GABA, 2 release DOPA, and 1 releases Glu

Striatum involves three nuclei

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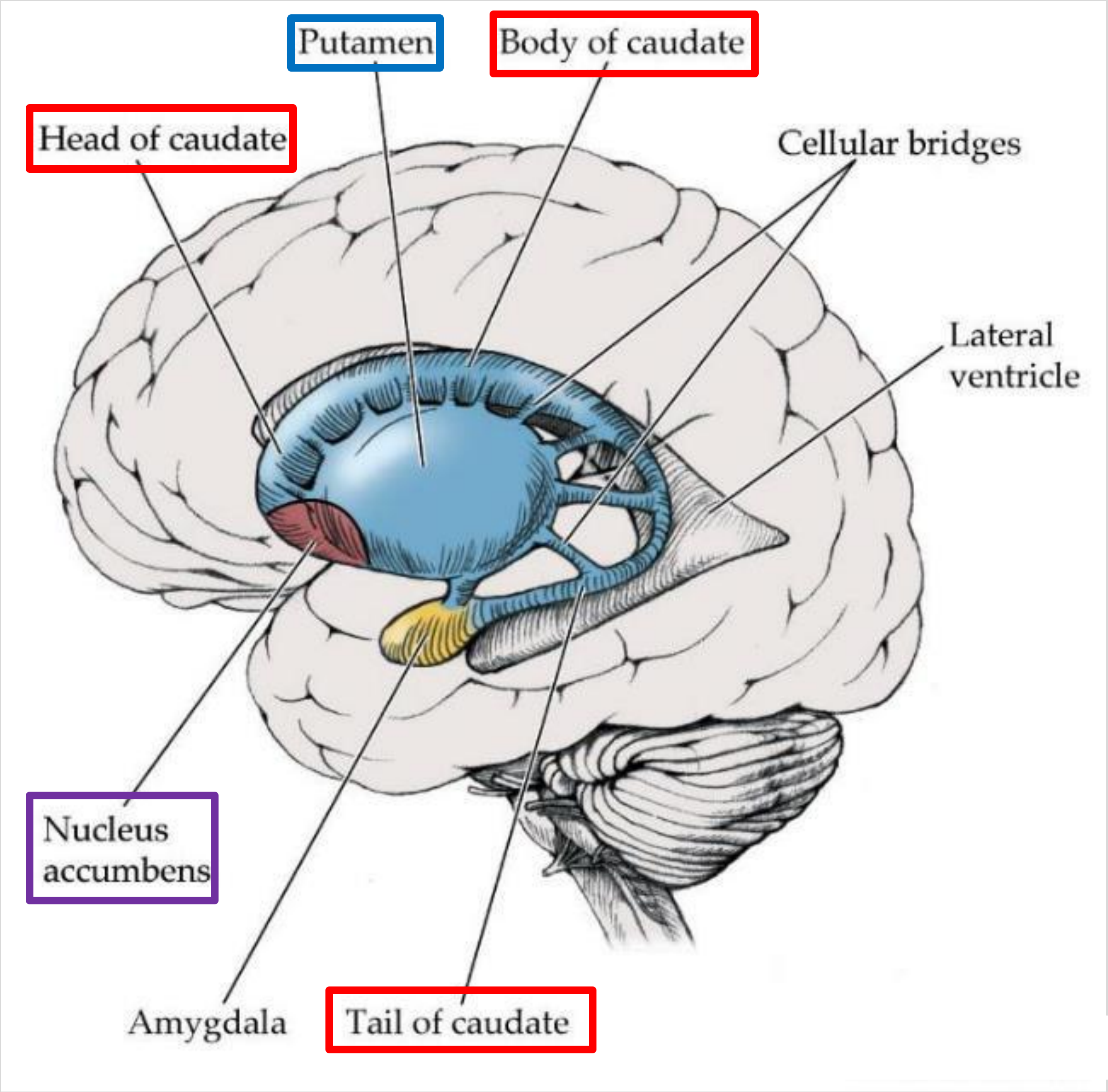

Neuroanatomy through clinical cases. Blumenfeld, 2nd edition

Caudate nucleus (dorsal striatum)
C-shaped structure (head body, and tail) that forms lateral walls of lateral ventricles.

Putamen (dorsal striatum)
Egg-shaped structure surrounded by the caudate

Grey matter connections between caudate and putamen separated by internal capsule fibers passing through. Looks like stripes, hence, “striated”.

Nucleus accumbens (ventral striatum)
Inferior to head of the caudate and anterior putamen

All are input nuclei and all release GABA.

Pallidum includes three nuclei

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Globus Pallidus*

Pale globe-shaped nucleus between the putamen and thalamus.

Separated from the thalamus by the internal capsule

(external segment, GPe)

- Division closest to the putamen
- Intrinsic nucleus, neither input nor output nucleus)

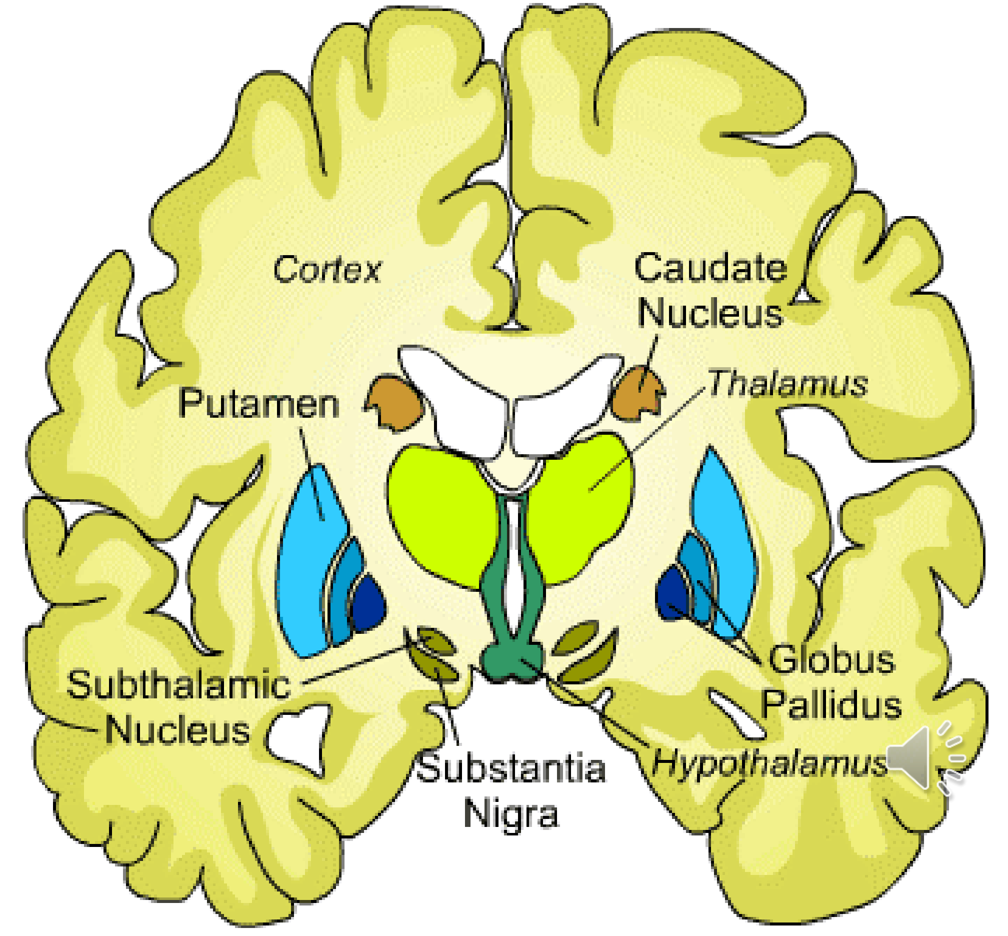
(internal segment, GPi)

- Division closest to the thalamus
- Output nucleus

Ventral pallidum

- Division closest to the thalamus
- Output nucleus

<u>**All pallidal nuclei release GABA.**</u>



*caudate + globus pallidus are called lentiform nucleus, looks like a bean

Nuclei within the diencephalon and midbrain

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Subthalamic nucleus (STN)

- Lateral to hypothalamus.
- Connected to both parts of globus pallidus.
- Input from cerebral cortex
- Releases glutamate

Substantia nigra (pars compacta)

- Inferior to subthalamic nucleus
- Modulatory nucleus (dopamine)

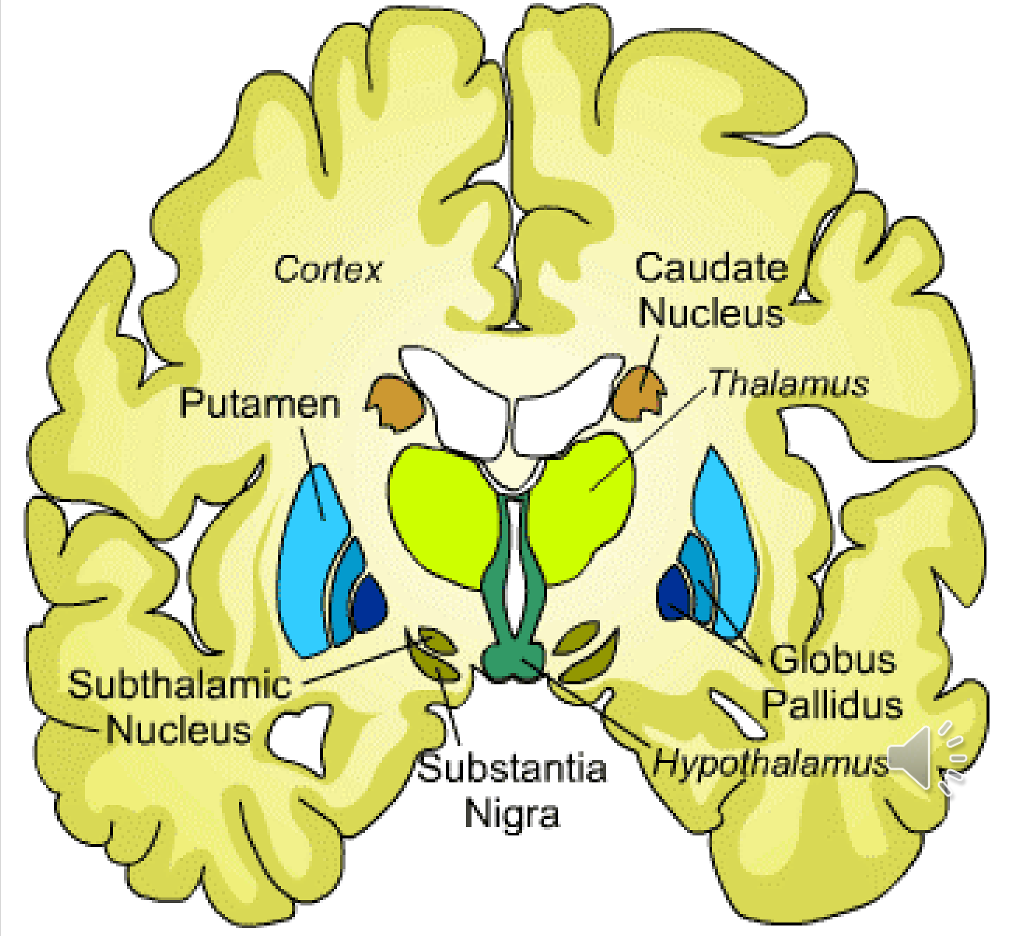
Substantia nigra (pars reticulata)

- Output nucleus
- Releases GABA

Ventral tegmental area (VTA)

- Medial to substantia nigra
- Modulatory nucleus (dopamine)

midbrain nuclei



Ten nuclei of the basal ganglia

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Striatum

Caudate – Input to BG, GABA
Putamen – Input to BG, GABA
Nucleus accumbens – Input to BG, GABA

Pallidum

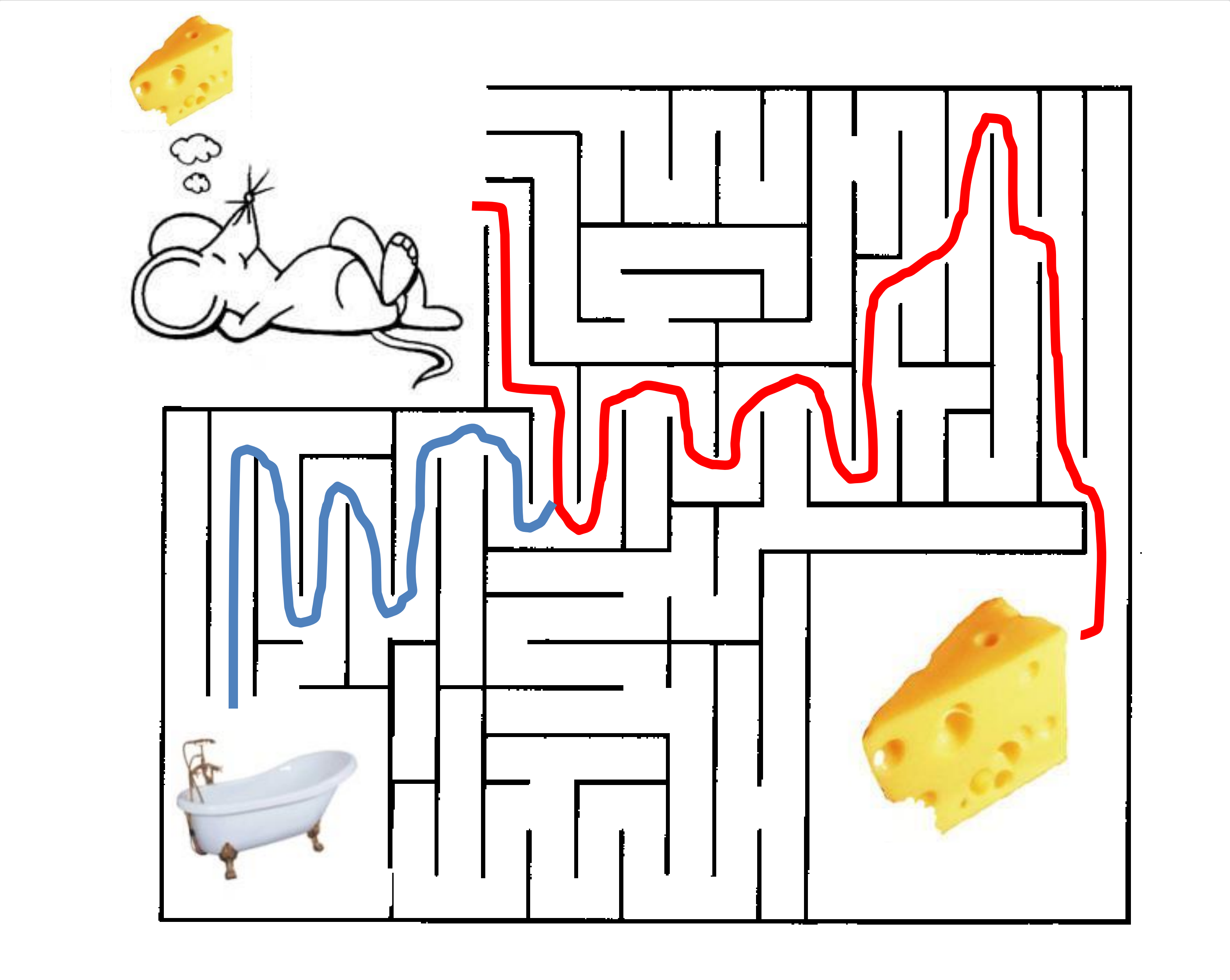
GPe – Intrinsic to BG, GABA
GPi – Output of BG, GABA
Ventral Pallidum – Output of BG, GABA

Nuclei within Midbrain and Diencephalon

Subthalamic nucleus – Input to BG+, Glu
Ventral tegmental area – modulation of BG, DA

Substantia nigra (pars reticulata) – Output of BG, GABA
Substantia nigra (pars compacta) – modulation of BG, DA

Complex relation between goal and actions



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<u>Goal</u>

Eat cheese

Take bath

<u>Possible actions</u>

Sniff

Turn right

Turn left

Walk forward

Stay still

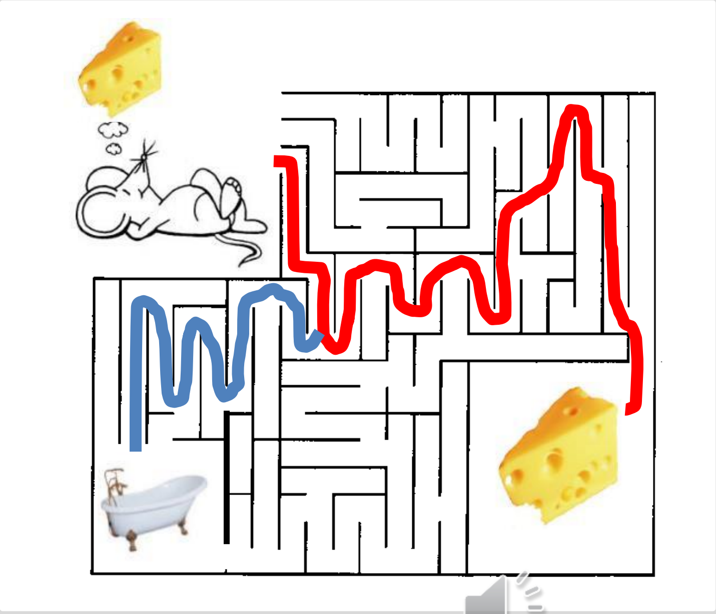
Clean paws

How to create a sequence of actions

Must promote a particular action, and demote other possibilities

Possible Actions	Sniff	Turn right	Turn left	Walk forward	Stay Still	Clean Paws
Possible Actions #1	Sniff	Turn right	Turn left	**Walk forward**	Stay Still	Clean Paws
Possible Actions #2	Sniff	**Turn right**	Turn left	Walk forward	Stay Still	Clean Paws
Possible Actions #3	Sniff	Turn right	Turn left	**Walk forward**	Stay Still	Clean Paws
Possible Actions #4	Sniff	Turn right	**Turn left**	Walk forward	Stay Still	Clean Paws

⋮



Sometimes need quickly halt a planned or ongoing action

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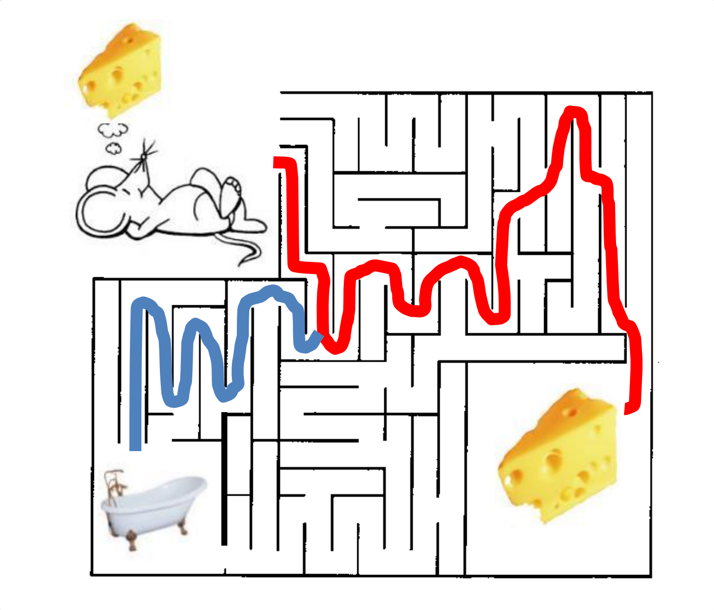


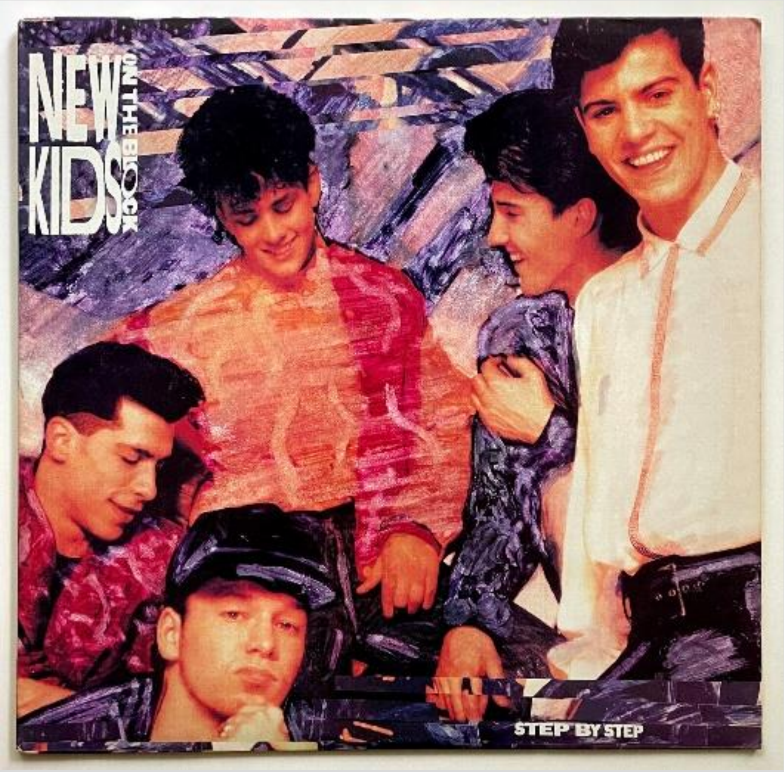
<u>Goal</u>
Eat cheese
Take bath

<u>Possible actions</u>
Sniff
Turn right
Turn left
Walk forward
Stay still
Clean paws

& Stop Everything!

Albin-DeLong Model

Proposed in 1990!



Most influential model of basal ganglia function to this day

Accounts for some key features of basal ganglia anatomy

Posits complementary pathways for promotion and demotion of potential actions:

the direct pathway and indirect pathway

Direct pathway promotes candidate actions

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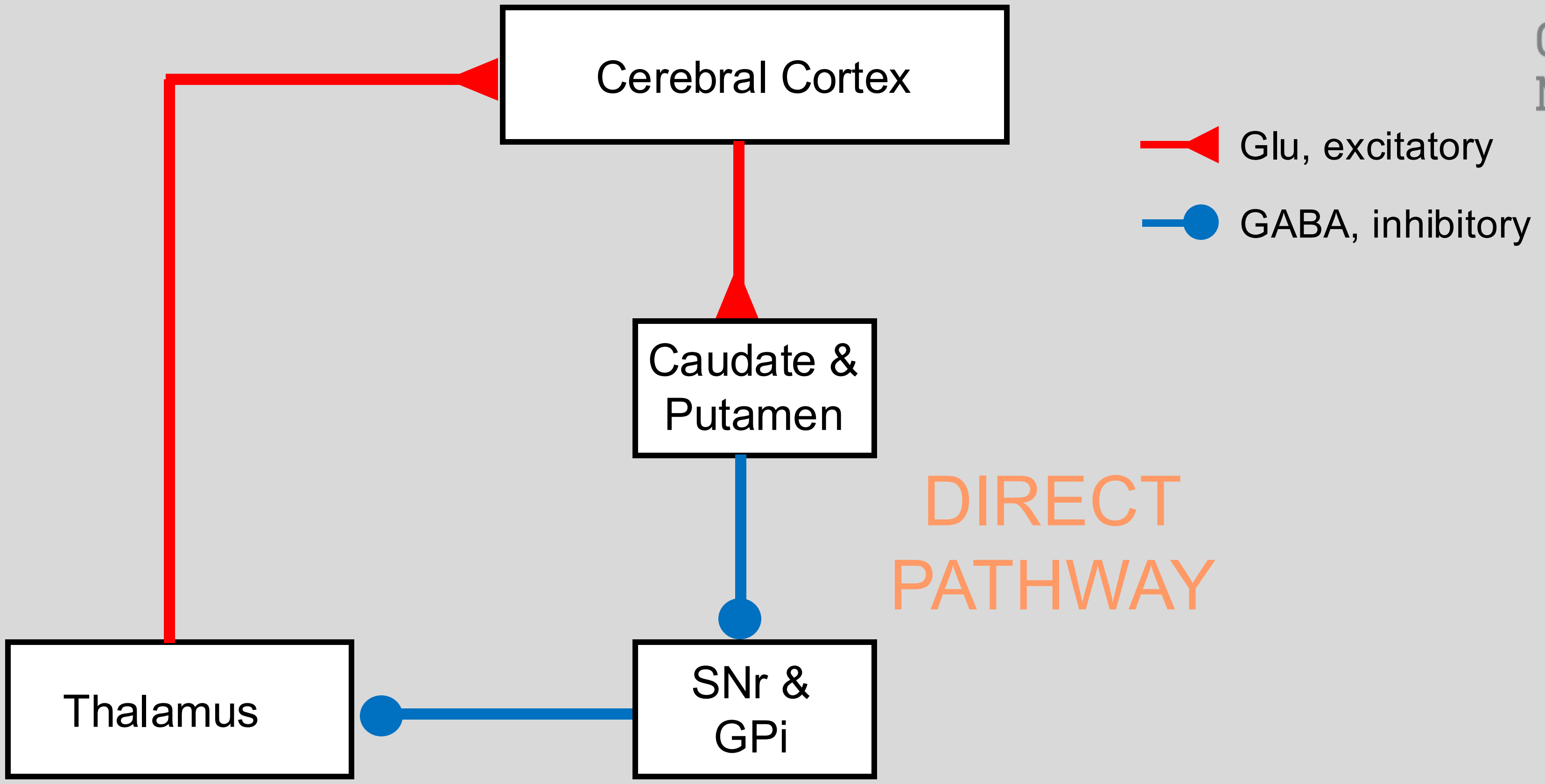

GPe – globus pallidus external
GPi – globus pallidus internal
STN – subthalamic nucleus
SNr – substantia nigra pars retic.

Excitatory cortical input projects to striate neurons in the caudate nucleus and putamen. Inhibitory neurons in the striatum project to inhibitory neurons in the globus pallidus (internal segment) which inhibits thalamus.

Disinhibition of thalamus leads to the initiation of an action.

Direct pathway promotes candidate actions

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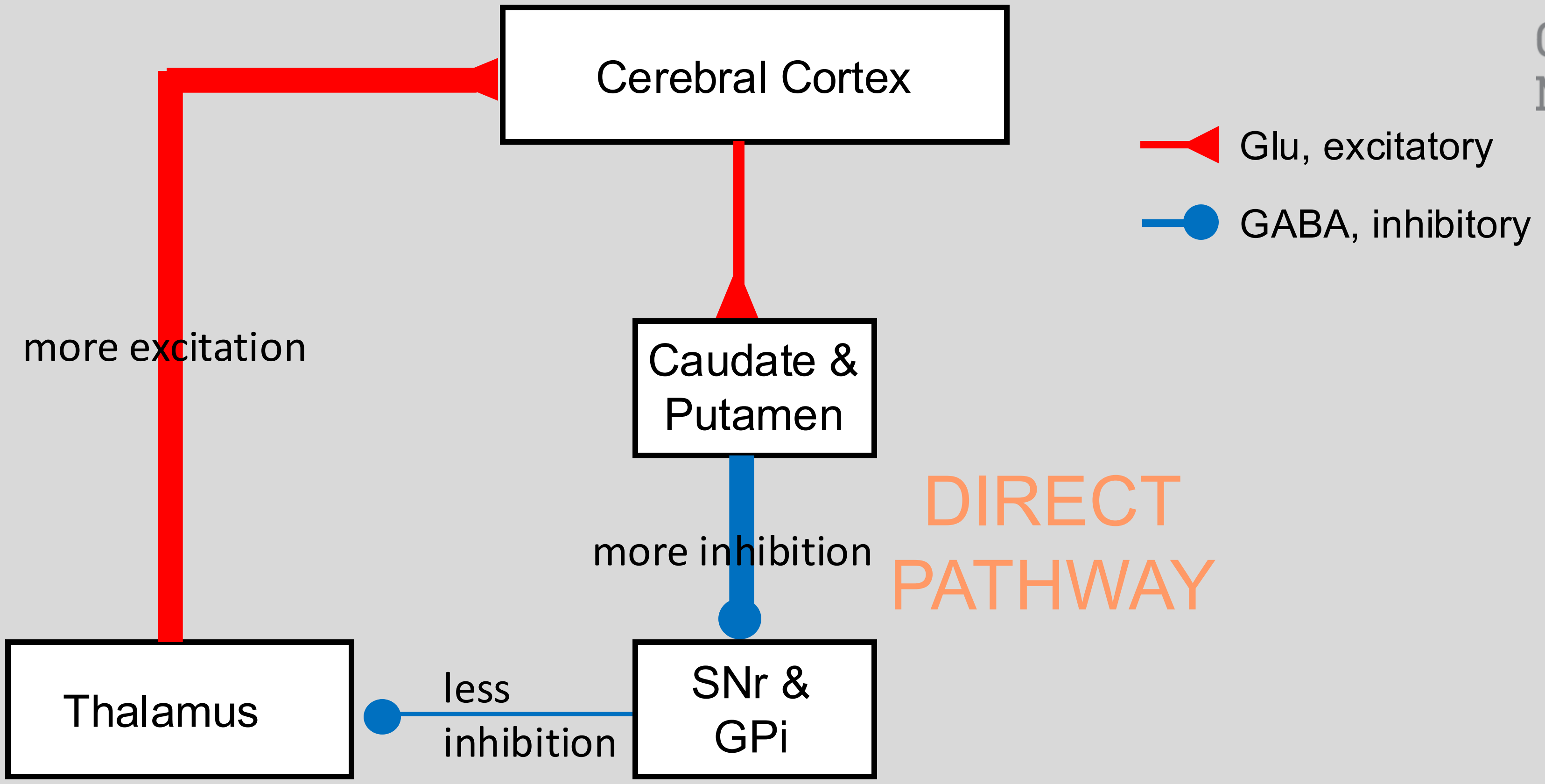

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<u>Disinhibition of thalamus leads to the initiation of an action.</u>

Indirect pathway demotes candidate actions

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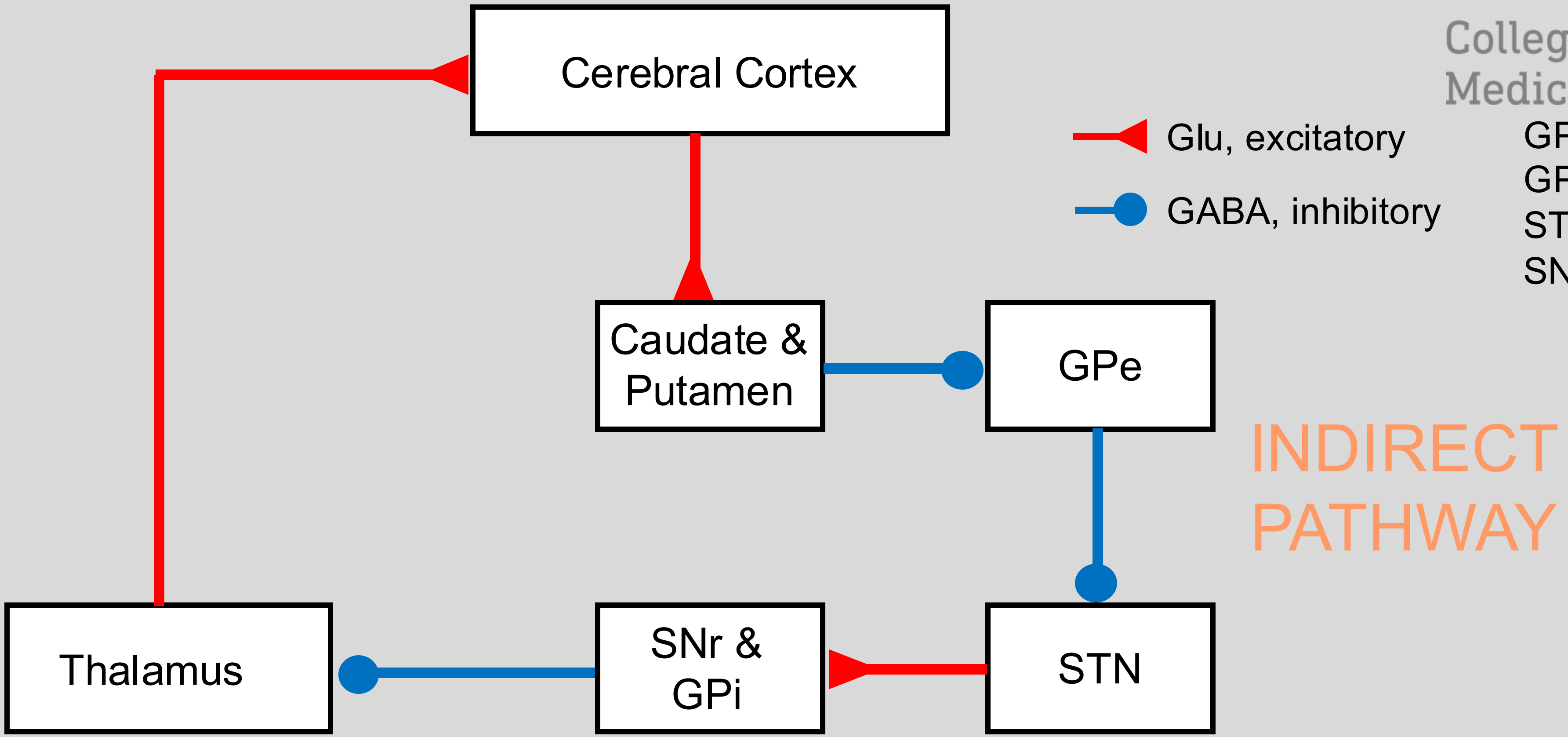

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STN excites Gpi. Disinhibition of STN leads to the inhibition of an action.

Indirect pathway demotes candidate actions

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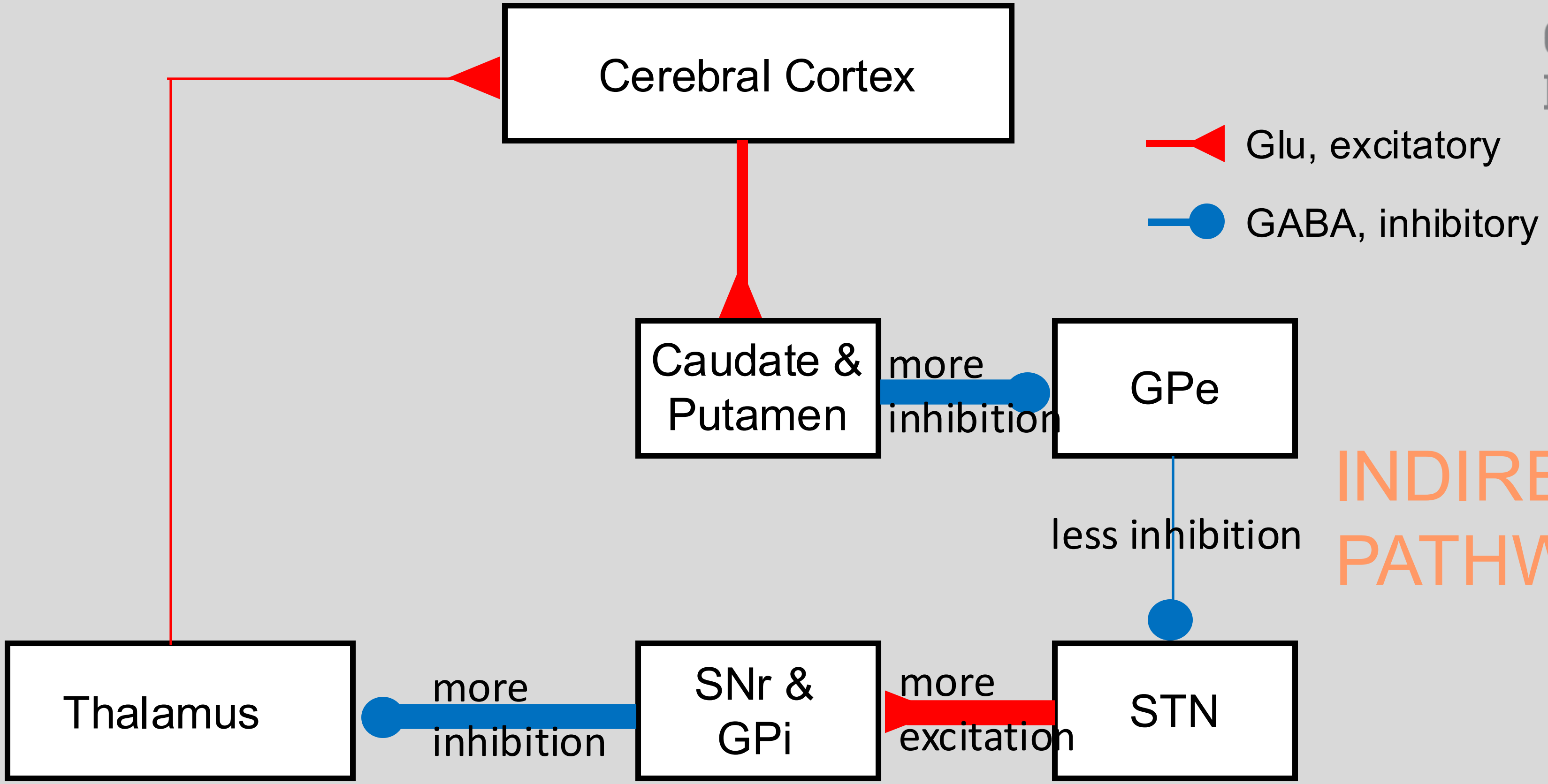

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INDIRECT PATHWAY

Excitatory cortical input projects to striate neurons in the caudate nucleus and putamen.
Inhibitory neurons in the striatum project to inhibitory neurons in the globus pallidus (external segment) which inhibits the subthalamic nucleus.
STN excites Gpi. Disinhibition of STN leads to the inhibition of an action.

Hyperdirect pathway: quickly blocks most/all candidate actions

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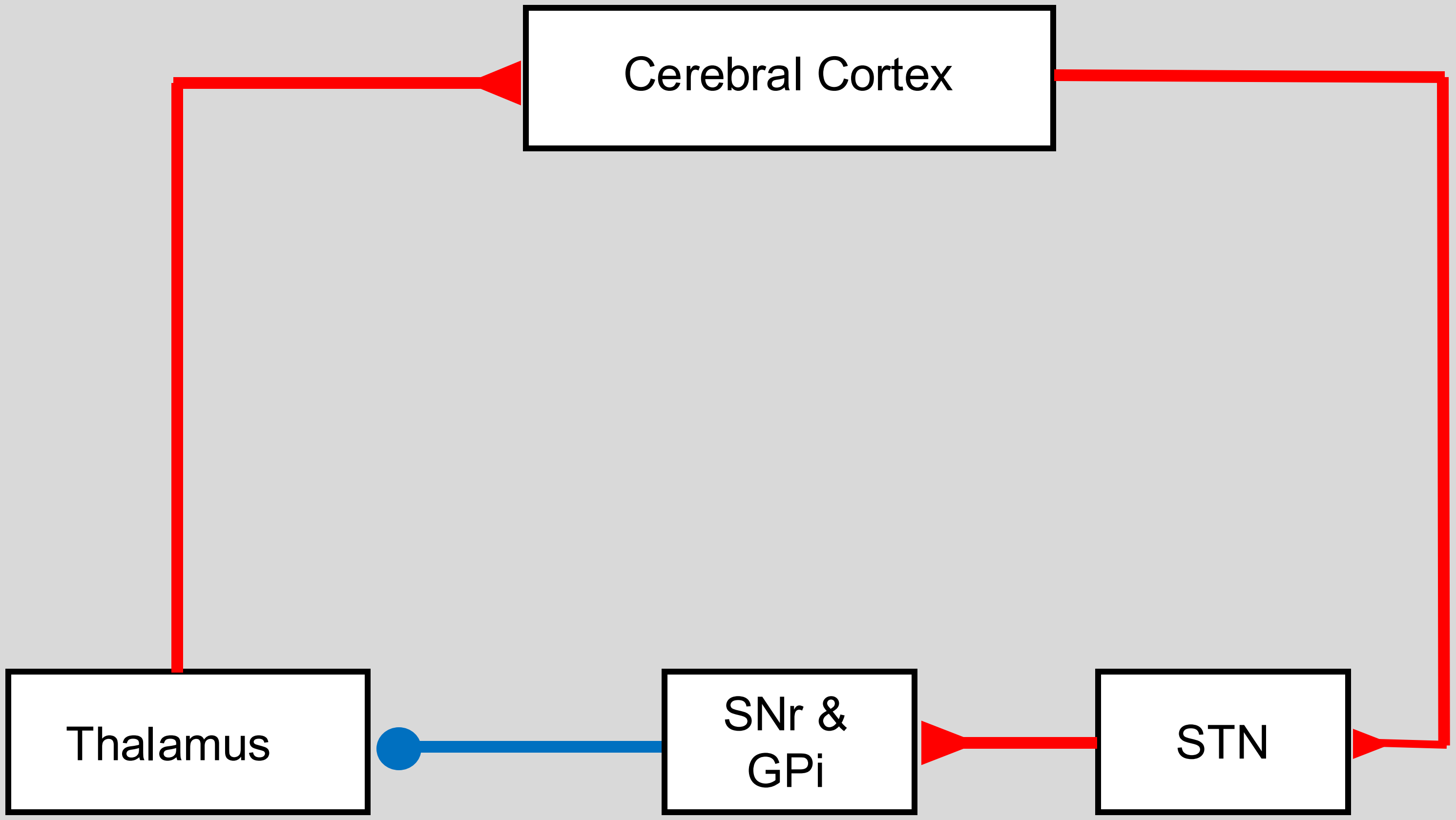
GPi – globus pallidus internal
STN – subthalamic nucleus
SNr – substantia nigra pars retic.



HYPERDIRECT PATHWAY

Excitatory cortical input to the subthalamic nucleus
STN excites GPi, and so leads to inhibition of thalamus.
Fastest pathway and affects the largest number of potential actions; countermands already initiated actions.

Impact of dopamine on the two pathways

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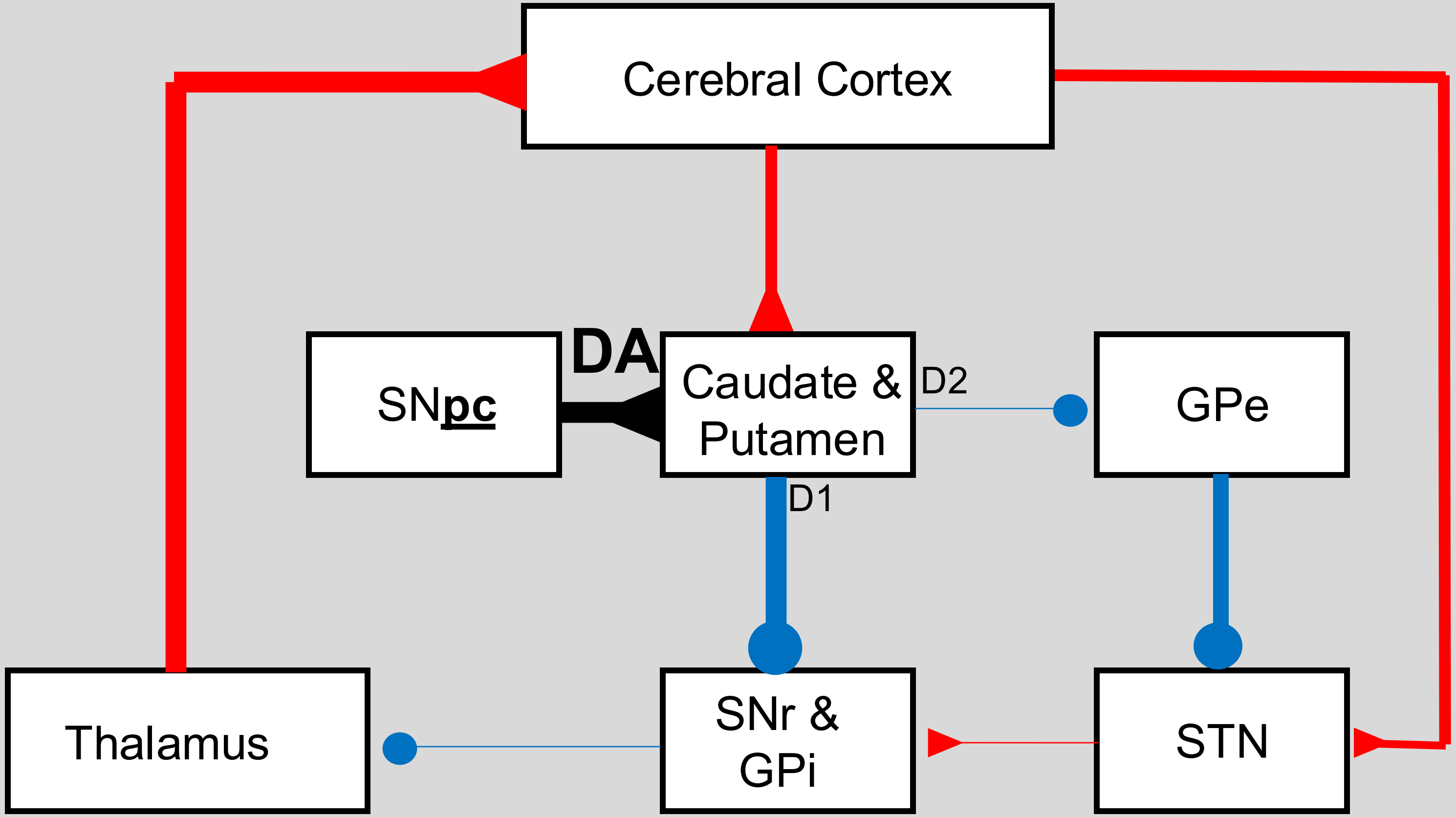

SNr – substantia nigra pars reticulata
SNpc – substantia nigra pars compacta

There are two classes of medium spiny neurons (MSN).
One class contains a D1 receptor for dopamine, the other a D2 receptor.
D1's G-coupled effect is to increase that neuron's excitability. More direct pathway activation
D2's G-coupled effect is to decrease that neuron's excitability. Less indirect pathway activation

accidentally misspeak on 1st pass

Opposite impact of Ach on the two pathways

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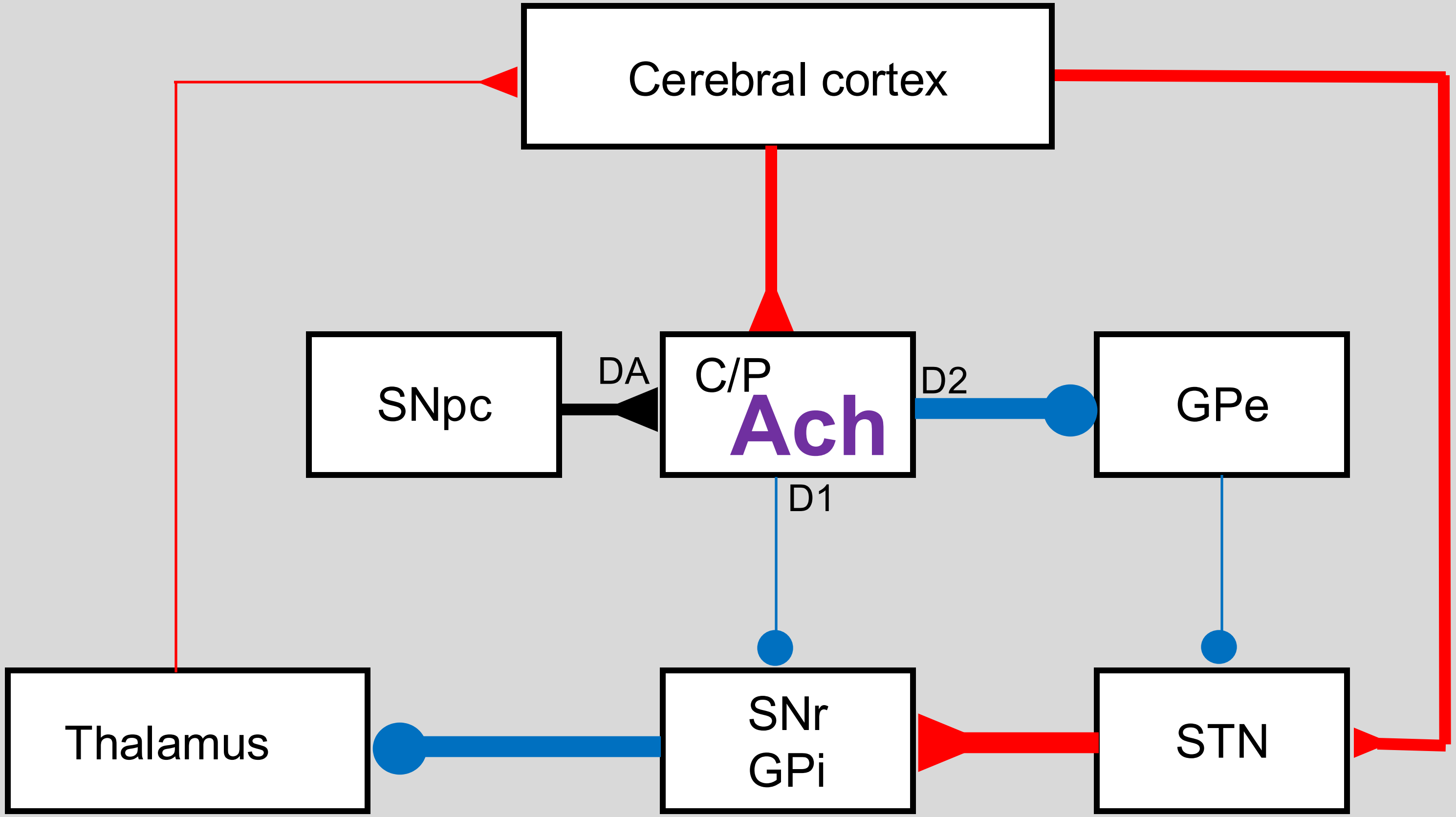
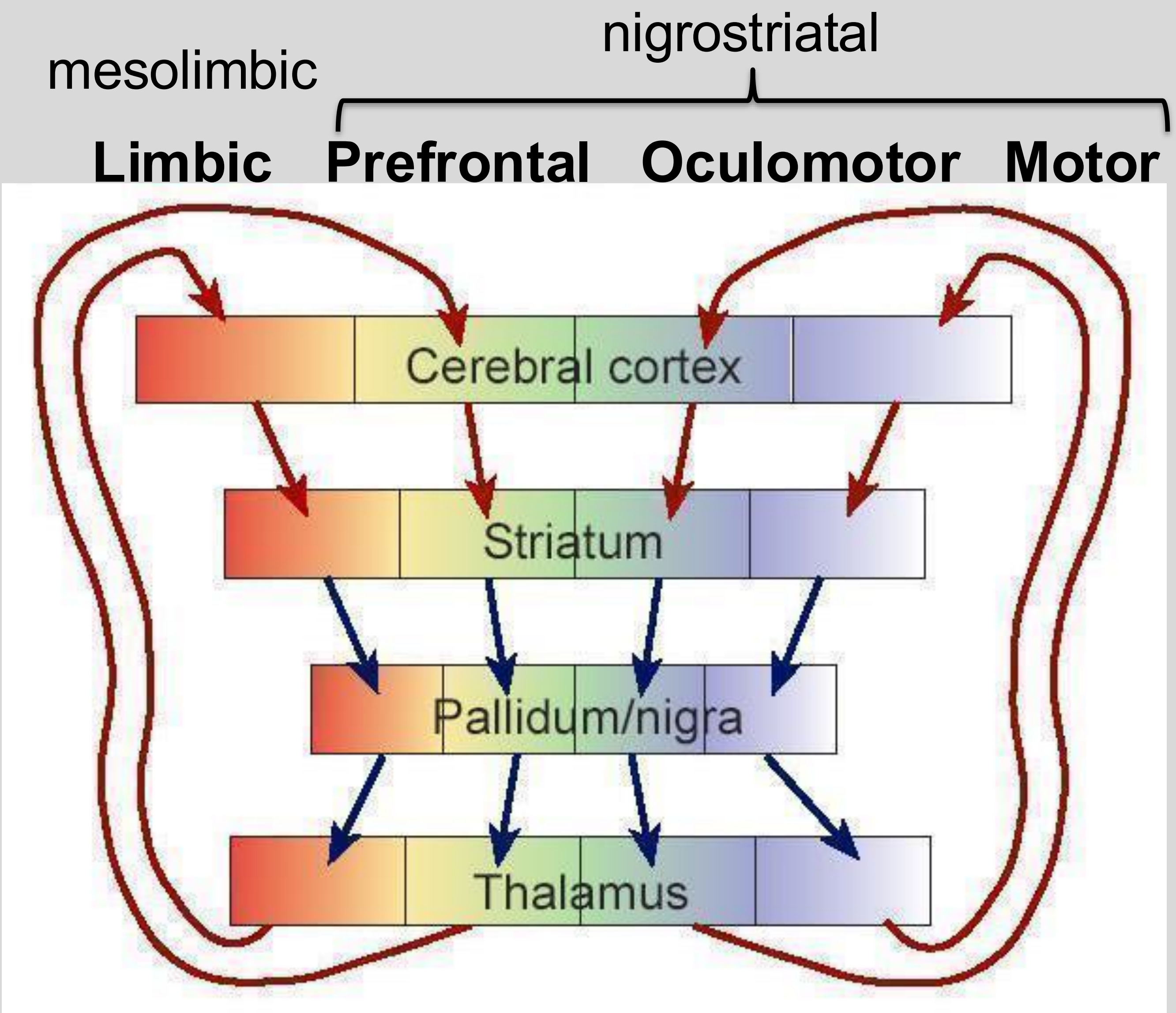


DA on D1: increase excitability in DIRECT pathway
DA on D2: decrease excitability in INDIRECT pathway

Ach on D1: decrease excitability in DIRECT pathway
Ach on D2: increase excitability in INDIRECT pathway

Opposite affects!

Distinct channels connect basal ganglia and cerebral cortex



Four channels involving basal ganglia, thalamus, and cerebral cortex

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Channel	Corticostriatal pathway	Striatal Nuclei	Pallidal Nuclei	Thalamic Nuclei	Cortical receiving areas
Motor	Primary somatosensory & motor areas; premotor and supplemental motor area	**Putamen**	**Globus pallidus; substantia nigra pars reticulata**	Ventral Lateral; Ventral Anterior	**Supplemental motor area, premotor and primary motor areas**
Oculomotor	Prefrontal and posterior parietal cortex	**Caudate nucleus**	**Globus pallidus; substantia nigra pars reticulata**	Ventral Anterior; MedioDorsal	**Frontal eye fields**
Prefrontal	Prefrontal and posterior parietal cortex	**Caudate nucleus**	**Globus pallidus; substantia nigra pars reticulata**	Ventral Anterior; MedioDorsal	**Prefrontal cortex**
Limbic	Temporal lobe; hippocampal formation; amygdala	**Nucleus accumbens**	**Ventral palladium**	MedioDorsal	**Cingulate gyrus and orbital frontal cortex**

The motor, oculomotor, and prefrontal channels all receive DA from the SNc.
The limbic channel receives DA from the ventral tegmental area.

Frontal cortical targets of the basal ganglia

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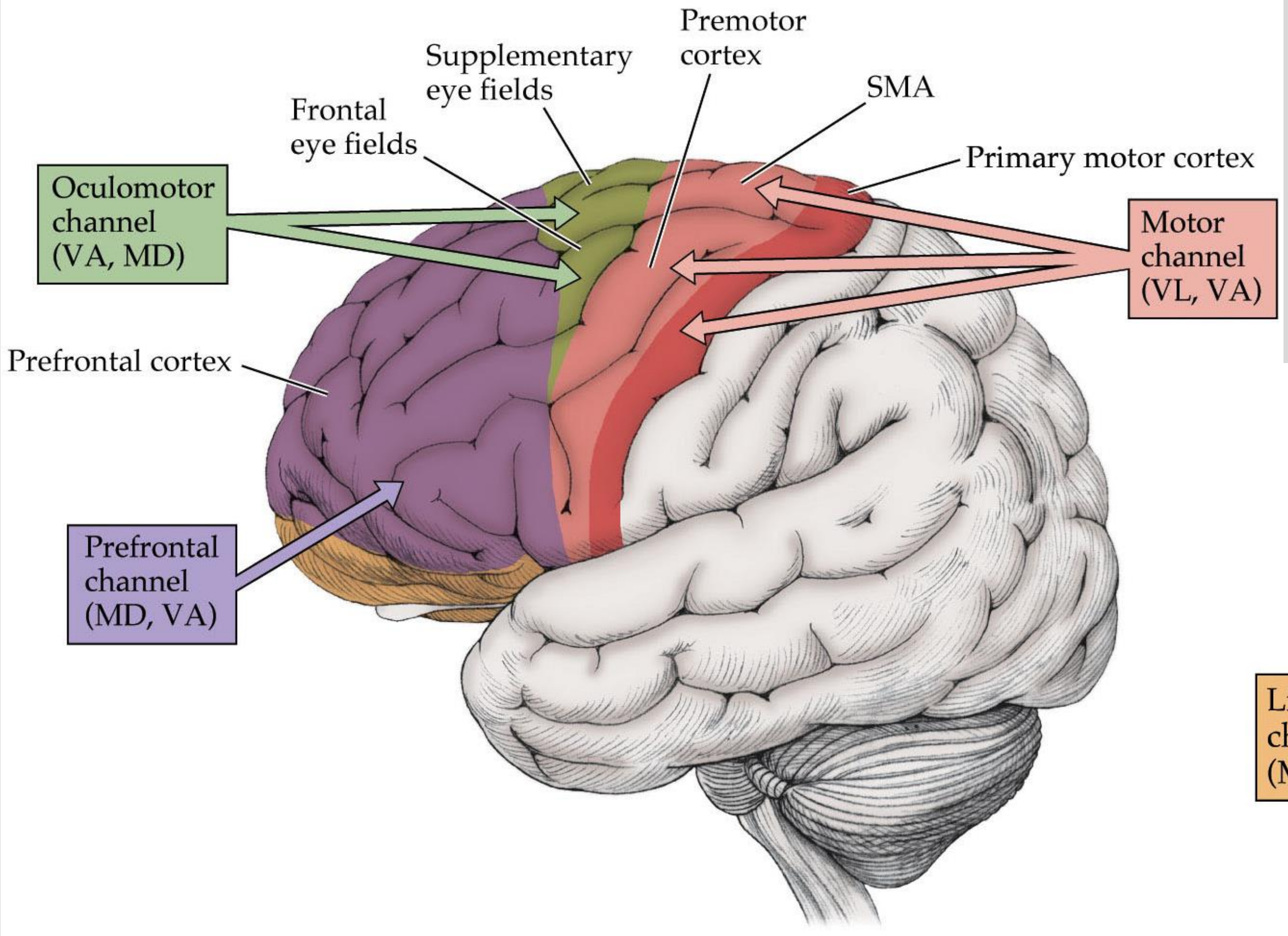

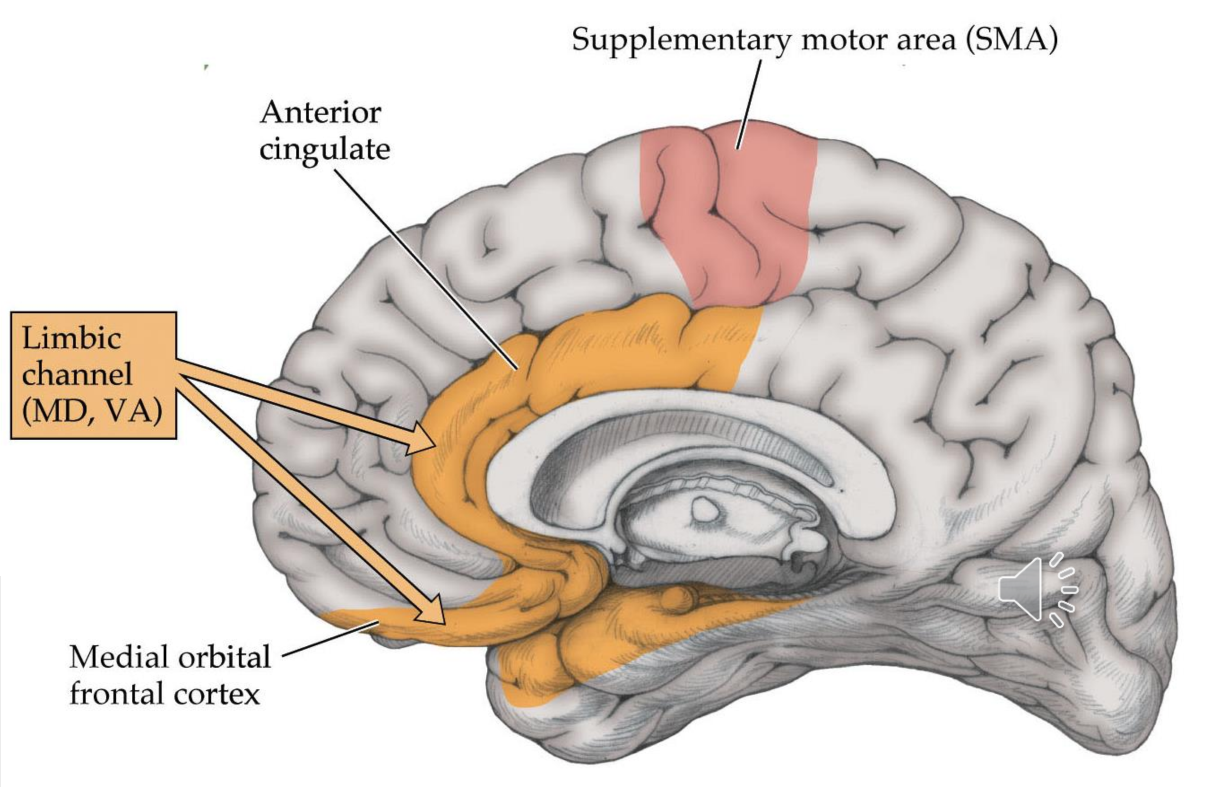

BG output also targets the brainstem

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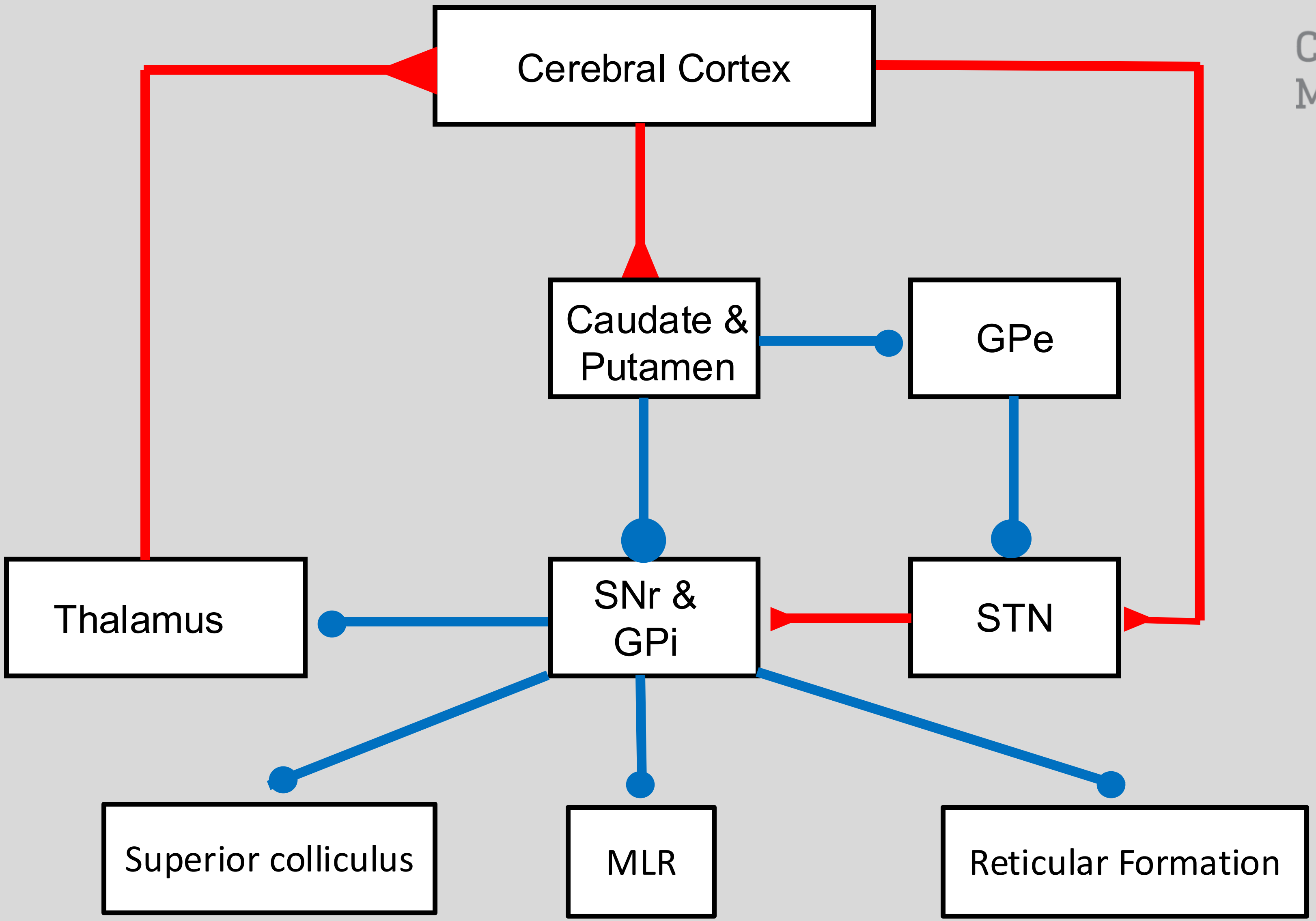

Ancient system

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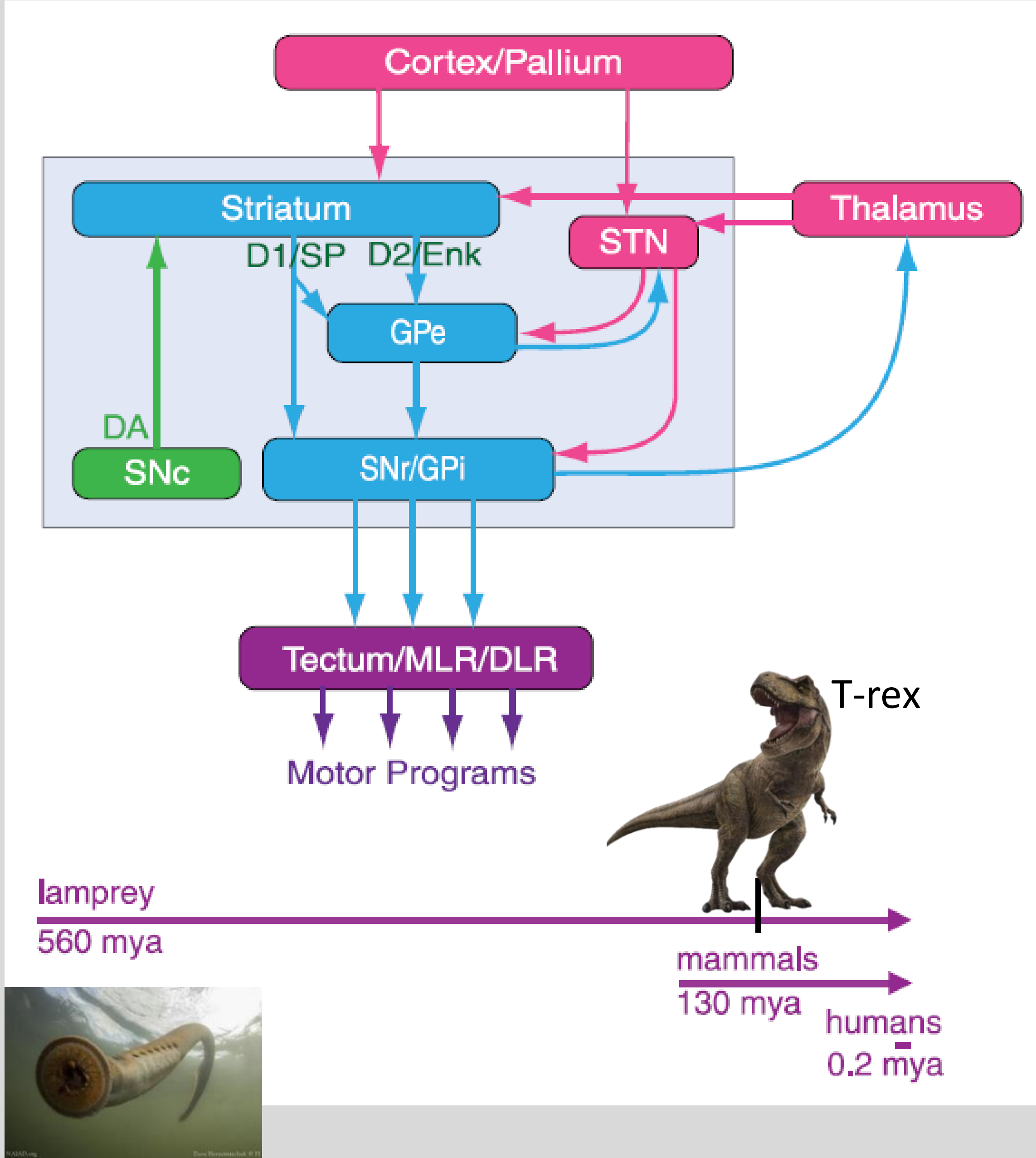

Grillner (2013) The evolutionary origin of the vertebrate basal ganglia and its role in action selection

Clinical manifestations of damage to basal ganglia

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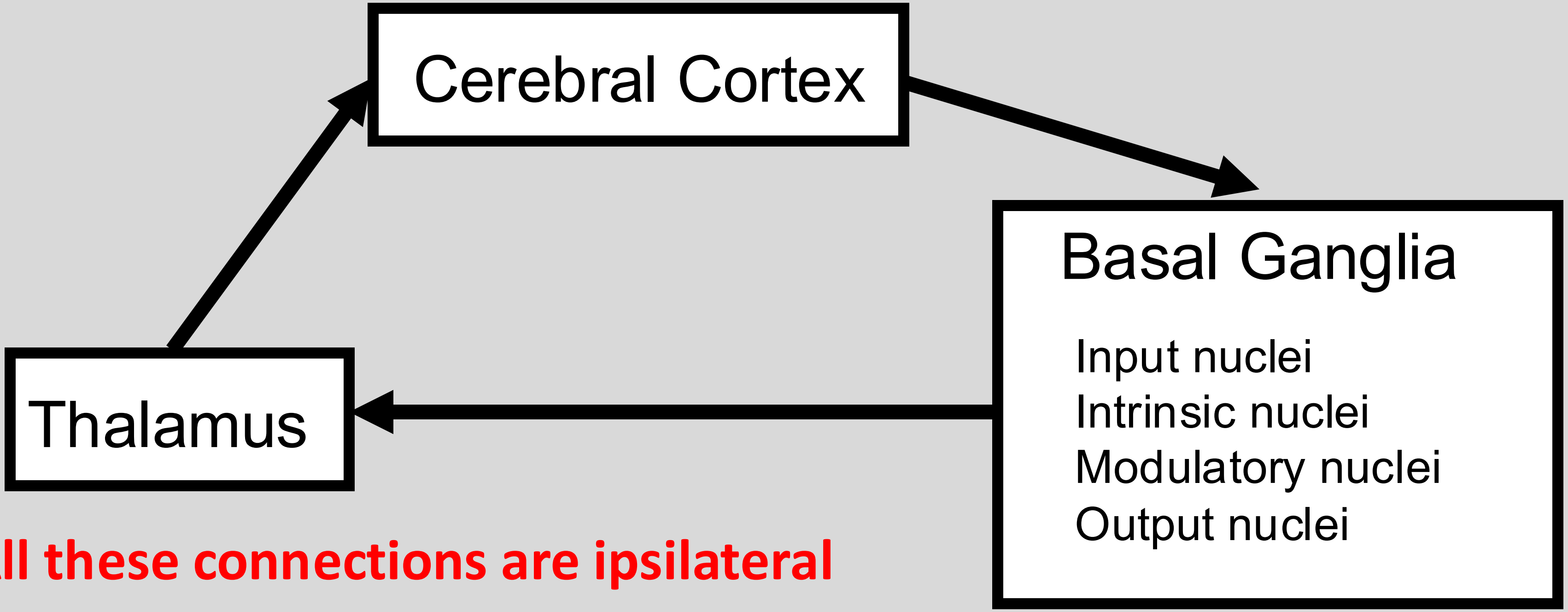
- Basal ganglia damage often presents with movement disorders.
- Movement disorders can be focal, generalized, unilateral or bilateral.
- In unilateral movement disorders caused by focal basal ganglia lesions, the movement disorder is contralateral to the lesion.



All these connections are ipsilateral

Too much or too little

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Hypokinetic disturbance

Slowed/absence of voluntary movement

Lesion to Direct Pathway

↓

Underactive cortex

Hyperkinetic disturbance

Numerous involuntary movements

Lesion to Indirect Pathway

↓

Overactive cortex

Parkinson's Disease

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Approximately 1% of the U.S. population over age 65

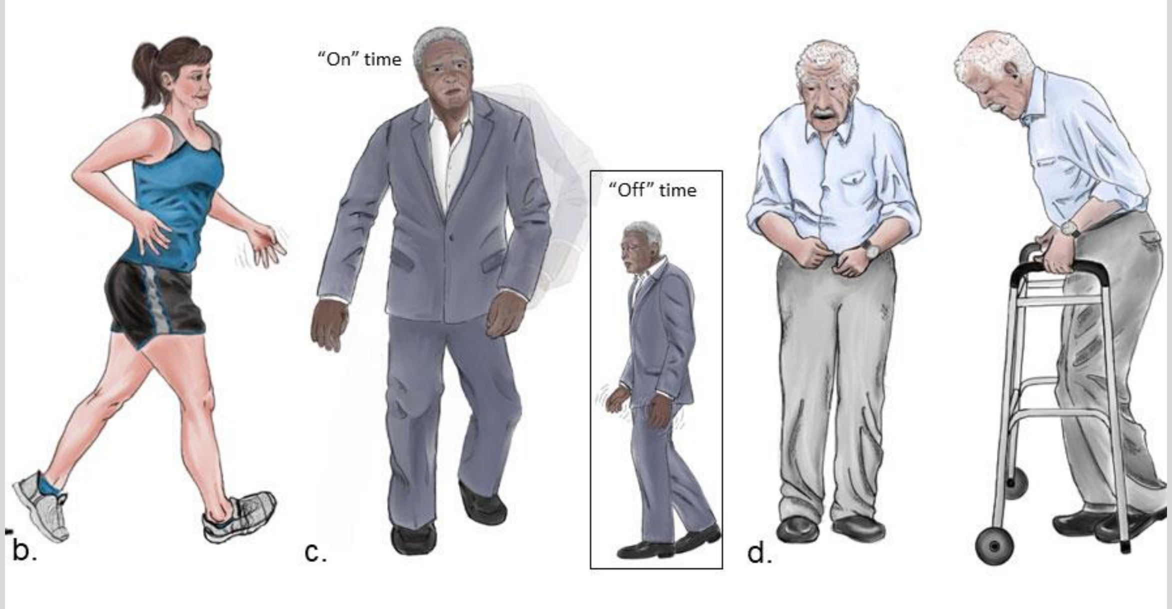
Usual age of onset is between 40-70 y/o

Generally no familial tendency

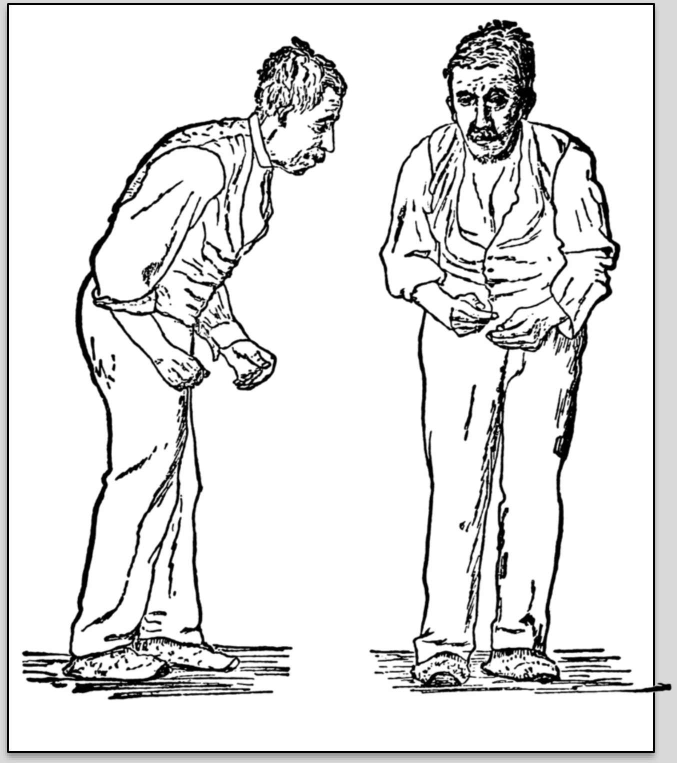
Typically idiopathic

Progression is insidious

Disability and death over the course of 10-30 years



Armstrong and Okun (2020)


Sir Willam Richard Gowers (1886)

Defining pathology of Parkinson's disease

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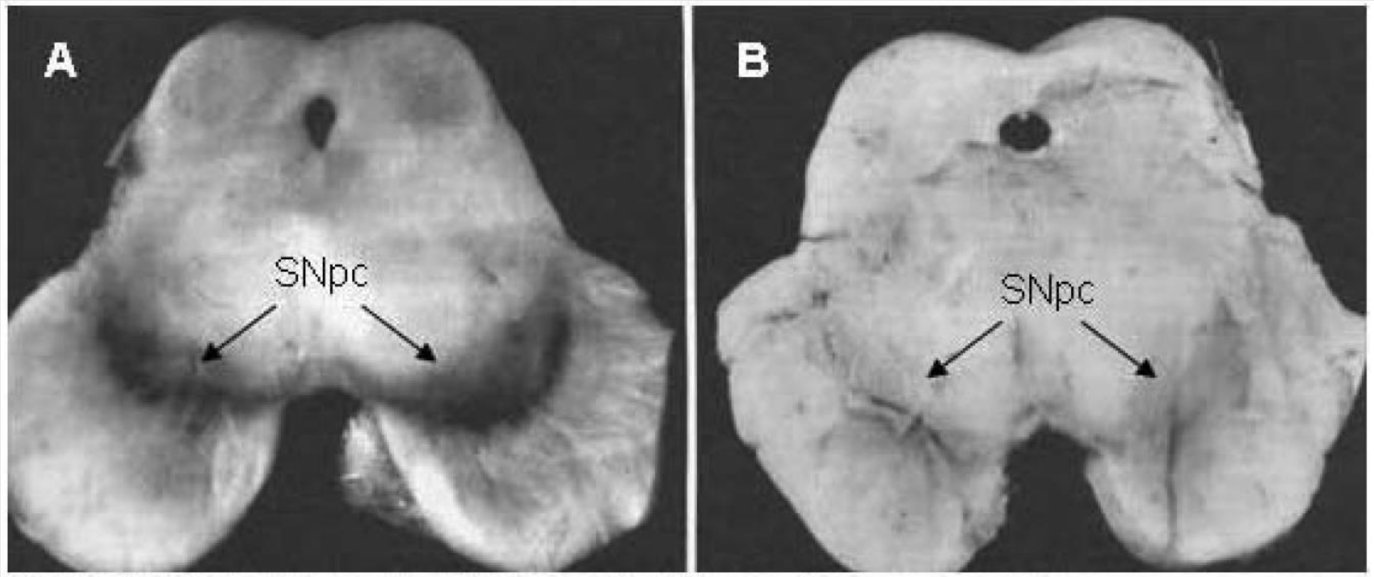
Massive loss of dopaminergic neurons in the Substantia Nigra (pars compacta), also in VTA

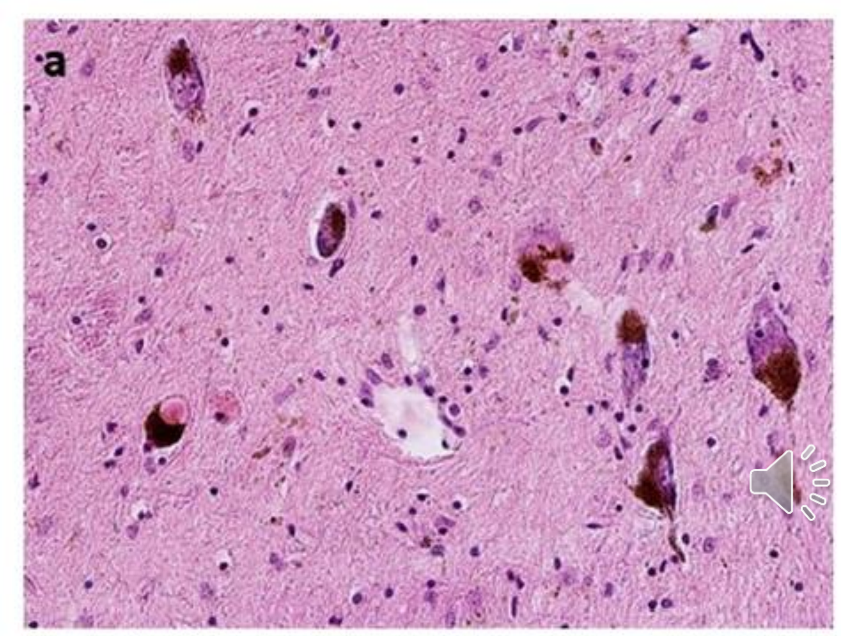


Lima (2012) Motor and non-motor features of Parkinson's disease

Neurons in distinct regions also contain cytoplasmic inclusions = Lewy Bodies

Contain ubiquitin, α-synuclein

Eosinophilic, Faint halo



Mensikova (2022) Lewy body disease or diseases with Lewy bodies?

Diagnosis of Parkinson's Disease

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Motor symptoms (TRAP)

- Tremor (<u>resting</u>)
- Rigidity
- Akinesia/Bradykinesia
- Postural instability

- Expressionless face
- Soft voice

https://www.youtube.com/watch?v=sJqKvajUC3k

Non-motor symptoms

- Anosmia – loss of smell
- Sleep disturbance – insomnia, fragmentation
- Autonomic – constipation, sialorrhea, excessive sweating
- Neuropsychiatric – mood disorder, apathy, dementia

<u>Non-motor symptoms often precede motor symptoms can be very disruptive</u>

Basal Ganglia Circuit: Parkinson's disease

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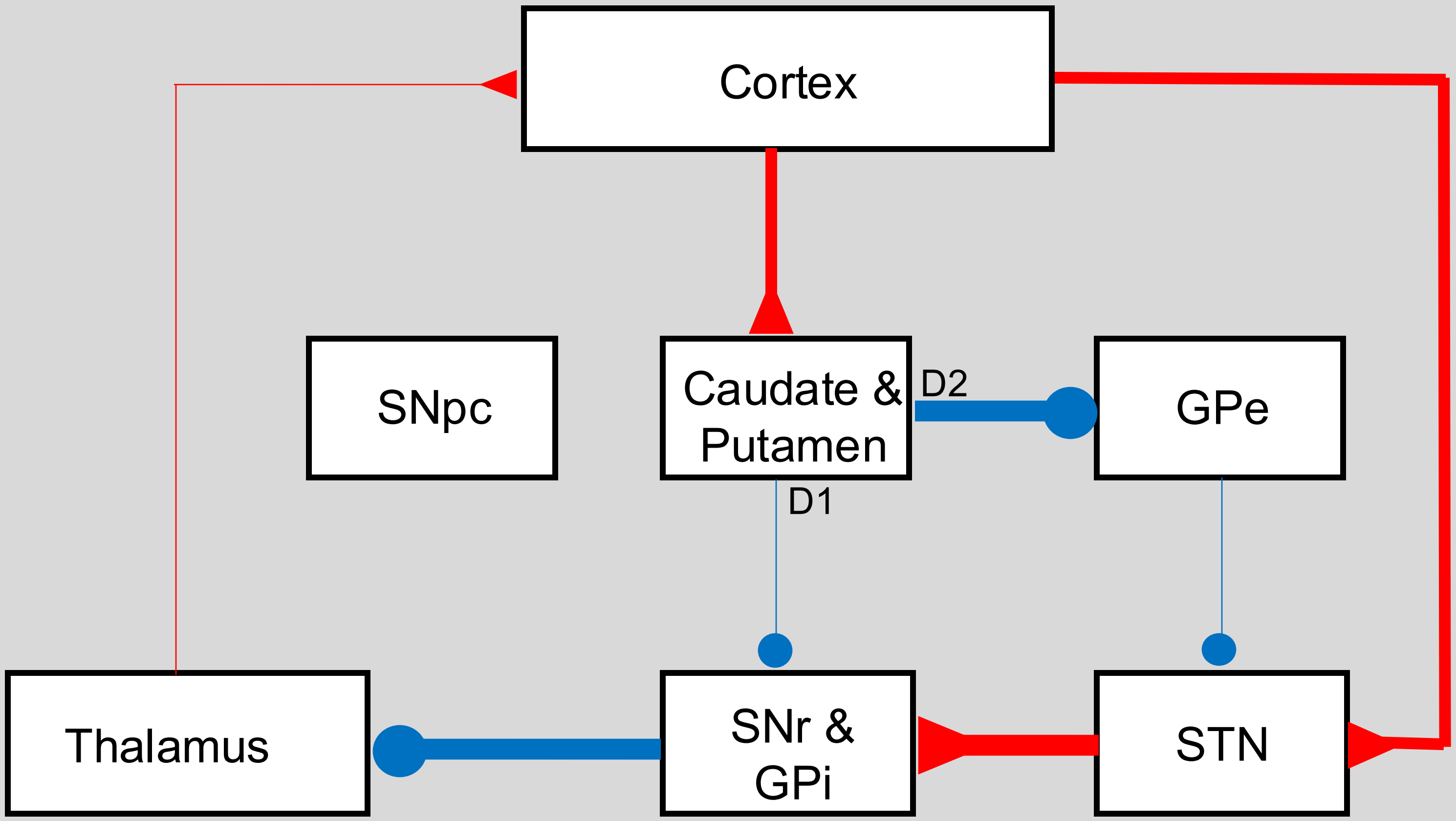

DA-D1: excitability ⇩ in DIRECT pathway to cortex
DA-D2: excitability ⇧ in INDIRECT pathway to cortex

DDx of Parkinsonism

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HEAD TRAUMA

- PD-like symptoms after various types of brain injury
- Particularly associated with prize fighting (Dementia pugilistica)

MANGANESE (used in steel production)

- Causes necrosis of the globus pallidus
- Toxins injure the brain in a way that produces PD-like symptoms.

MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine)

- MPPP is a opioid analgesic drug
- Incorrect production created MPTP→ crosses BBB.
- Neurotoxic effects in substantia nigra pars compacta
- “The case of the frozen addicts”
- Used in animal models of Parkinson’s

A diagnosis other than idiopathic PD should be suspected if symptoms are symmetric, absence of resting tremor, or lack of response to DA-agents

Treatment of PD

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Pharmacologically restore or aurgment dopamine

- Dopamine precursor: Levodopa
- D2 receptor agonist: Bromocriptine

Pharmacologically decrease Ach activity

- Benztropin

Surgical lesion

- Pallidotomy

Virtual lesion

- Deep brain stimulation

All treatments impact symptoms, do not cure underlying cause

Hemiballismus

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Lesion of the subthalamic nucleus

Results in motor dysfunction <u>of one side of the body</u>

Spontaneous involuntary movements with rapid onset and speed, i.e. ballistic

Typically caused by local vascular occlusion

-Decreased pallidal inhibition of thalamus (remember hyper-direct pathway!)

Movements can be improved with dopaminergic antagonists, e.g. Haloperidol

https://www.youtube.com/watch?v=RAATfknm1F4

Huntington’s Disease

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- Inherited autosomal dominant disease.
- Expansion of CAG repeats for encoding the protein huntingtin

  Normal individuals have <30 repeats

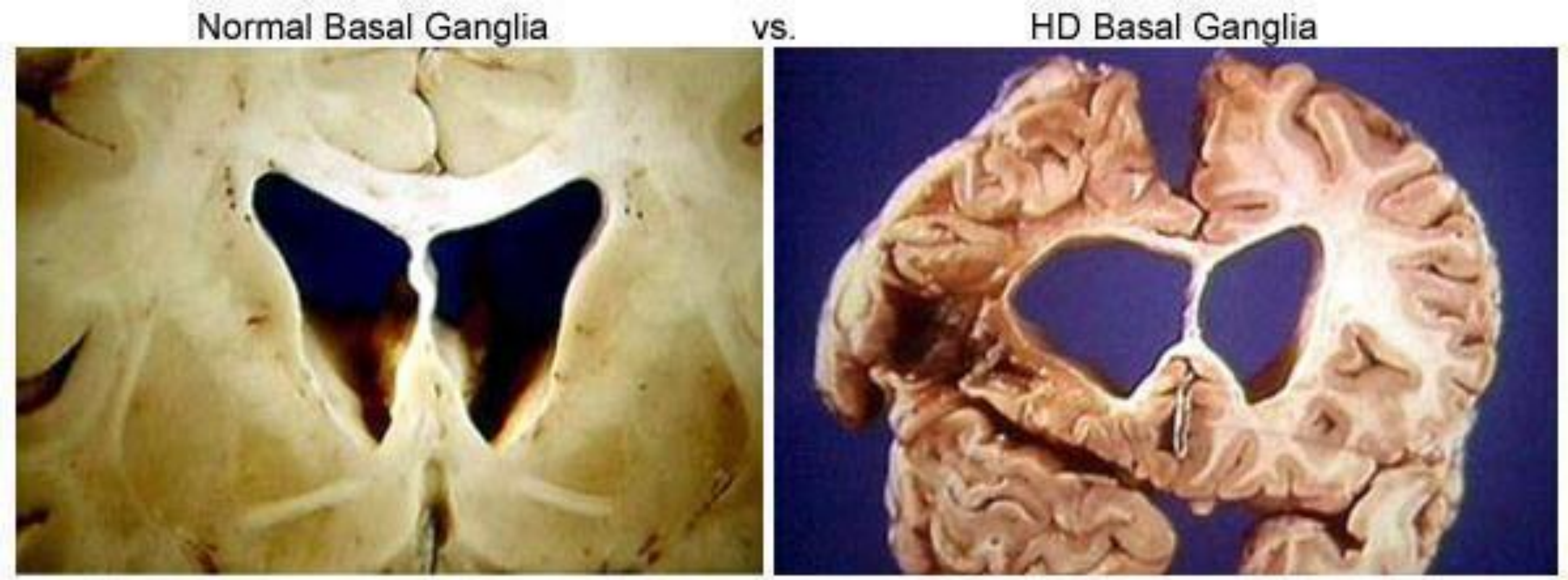
  Individuals with >40 repeats have HD or will develop symptoms

  Usually presents in 30-40s
- Progressive deterioration of motor function (chorea: rapid transient dance-like movements) and mental function (psychosis)
- Death ~10-20yrs
- No effective treatment for the progressive dementia; Anti-dopaminergic medication for motor symptoms.

https://www.youtube.com/watch?v=RxWEilu-Mf4

Huntington’s Disease

Atrophy of striatum: caudate nucleus, putamen, and nucleus accumbens
Disintegration of input nuclei of basal ganglia
Dilation of lateral and 3rd ventricles



The basal ganglia of the human brain, showing the impact of HD on brain structure in this region. Note especially that the brain of a person with HD has bigger openings due to the death of nerve cells in that region.

Source: Singer, Jonathan. Huntington's Disease. Online. Available at: http://ist-socrates.berkeley.edu/~jmp/HD.html

Treatments for different disorders can induce PD-like symptoms

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D2 antagonists - antipsychotics and anti-emetics

- Rigidity, hypokinesia, resting tremor
- Abrupt onset, symmetrical symptoms
- Symptoms can persist for weeks after removal of offending agent, and sometimes permanent symptoms

"Tardive dyskinesia" – ("tardy" means "late")

- These syndromes present after a latency period following initiation of medications, > 3 months
- Often repetitive stereotypical movements of the mouth and tongue such are chewing and smacking of the mouth and lips, tongue rolling or pushing against the inside of the cheek.
- Can be voluntarily suppressed, but returns upon distraction.

Summary

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- Basal ganglia is collection of nuclei communicating with cerebral cortex
- Disinhibition is a prominent feature of internal processing
- Three primary circuits in Albin-DeLong model: direct pathway (promoting actions), indirect pathway (demoting actions), and hyper direct pathway (global suppression of actions).
- Some features of basal ganglia dysfunction can be understood with this model

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R